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Construction of a predictive model for retinopathy of prematurity using machine learning algorithms

Yong Yang[1], Lili Li[2], Yu Li[3], Linfeng Song[1], Huiyi Zuo[1], Renshen Shi[1], Peng Fu[4] and Minli Huang[1*]

Abstract

Background Retinopathy of prematurity (ROP) has emerged as one of the leading causes of visual impairment or blindness among newborn infants worldwide. The purpose of this study was to develop a predictive model for ROP using machine learning methods.

Methods A retrospective study was conducted on 586 neonates admitted to the Department of Neonatology at the First Affiliated Hospital of Guangxi Medical University from January 2019 to January 2024, who met the inclusion criteria and underwent ROP screening.1.ROP-related risk factors were collected by reviewing electronic medical records during hospitalization and follow-up outpatient visits.2.Lasso regression was applied to screen ROP-related risk factors, identifying significant predictors. Seven machine learning models were constructed using these predictors. Model performance was evaluated and compared based on metrics including Area Under the ROC Curve (AUC), accuracy, precision, sensitivity, specificity, F1-score, and Kappa coefficient.

Results 1. Lasso regression screened 109 ROP-related risk factors and identified 46 significant predictors. These factors were used to construct seven machine learning models.2. Among the models, the random forest (RF) algorithm demonstrated optimal performance, with the following metrics: Training set: AUC: 1.000; accuracy: 99.7%; precision: 99.7%; specificity: 99.7%; Sensitivity: 99.7%; F1-score: 0.997; Kappa coefficient: 0.994.Testing set: AUC: 0.981; accuracy: 95.7%; precision: 92.3%; specificity: 99.3%; Sensitivity: 66.7%; F1-score: 0.774; Kappa coefficient: 0.751.

Conclusion The RF predictive model based on 46 significant ROP-related risk factors exhibits strong predictive value for ROP occurrence. This model provides a useful tool for early clinical identification of high-risk ROP populations.

Keywords Retinopathy of prematurity, Machine learning, Predictive model, Lasso regression

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Introduction

ROP, a developmental retinal vasoproliferative disorder affecting preterm and low-birth-weight infants [1], has emerged as a leading cause of visual impairment and blindness in newborns globally [2]. Characterized by rapid progression and a narrow therapeutic window, ROP carries an extremely high risk of blindness once it advances to late-stage tractional retinal detachment, significantly compromising long-term quality of life [3]. As an avoidable blinding disease, early screening, accurate diagnosis, and timely treatment for high-risk preterm infants can effectively reduce ROP-related blindness [4]. However, fewer than 5% of screened infants ultimately require intervention [5], while repetitive and inefficient screening for non-treatment-eligible cases not only strains healthcare resources but also imposes psychological trauma on infants and financial burdens on families [6]. Given its clinical occult nature and progression risks, establishing a clinical predictive model-based early screening system has become pivotal for prevention and control [7].

In recent years, numerous researchers worldwide have developed predictive models for identifying high-risk neonates requiring ROP treatment, aiming to optimize screening efficiency by reducing unnecessary examinations. These ROP prediction models extend beyond conventional gestational age (GA) and birth weight (BW) parameters by incorporating additional clinical indicators: The Utrecht model [8]: incorporates GA, BW, and number of red blood cell transfusions during the first 4 postnatal weeks, The PINT model [9]: utilizes GA, BW, and daily weight gain rate. However, significant limitations persist in current models. First, their generalizability is constrained by regional variations in ROP incidence rates, disease progression patterns, and treatment protocols. Second, most existing models rely on traditional logistic regression analysis, which demonstrates limited predictive power when handling multivariable interactions and nonlinear relationships [10]. To address these limitations, this study proposes a machine learning approach for constructing predictive models, thereby offering a novel methodology for optimizing ROP screening protocols.

Materials and methods

Study population

A retrospective study was conducted on 586 infants who underwent ROP screening in the Neonatal Department of the First Affiliated Hospital of Guangxi Medical University from January 2019 to January 2024. The cohort included 262 females (44.70%) and 324 males (55.30%). The gestational age ranged from 23 to 35 weeks (mean 31.51 ± 1.82 weeks), and birth weight ranged from 0.39 kg to 3.35 kg (mean 1.80 ± 0.46 kg). Among these infants, 4 (0.70%) were small for gestational age (SGA), and 87 (14.84%) were twins.

Inclusion Criteria

Eligible infants met the screening criteria outlined in the *Chinese Guidelines for Retinopathy of Prematurity Screening (2014)* [11] and the *Expert Consensus on Classification and Management of Retinopathy of Prematurity in China (2023)* [12]. Specifically:1. Preterm infants with gestational age < 32 weeks or birth weight < 2000 g;2. High-risk infants identified by pediatricians due to severe comorbidities or prolonged oxygen therapy.

Exclusion Criteria

1. Incomplete clinical data;2. Severe congenital malformations or chromosomal abnormalities;3. Ocular pathologies affecting screening (e.g., congenital cataract, glaucoma, intraocular tumors, extensive retinal hemorrhage, familial exudative vitreoretinopathy, congenital corneal opacity); 4. Loss to follow-up before achieving retinal vascularization or regression as per guidelines.

Research methods

Screening criteria

The initial examination should commence 4–6 weeks after birth or when the corrected gestational age (CGA) reaches 31–32 weeks. The screening intervals are as follows: (1) Zone I: No ROP, Stage 1, or Stage 2 ROP: weekly examination; (2) Zone I regressing ROP: examination every 1–2 weeks; (3) Zone II Stage 2 or 3 ROP: weekly examination; (4) Zone II Stage 1 ROP: examination every 1–2 weeks; (5) Zone II Stage 1 or no ROP, Zone III Stage 1 or 2 ROP: follow-up every 2–3 weeks.

Termination criteria (meeting any one of the following conditions): (1) Retinal vascularization (nasal side reaching the ora serrata, temporal side within 1 optic disc diameter of the ora serrata); (2) CGA ≥ 45 weeks with no prethreshold or threshold ROP, and retinal vascularization extending to Zone III; (3) Regression of retinopathy.

Interpretation of screening results

The most severe ROP condition observed in each infant was recorded as the basis for classification. ROP lesions were categorized by zone, stage, and presence of plus disease according to the International Classification of ROP (ICROP) [7]. Type1 ROP: Any stage of ROP in zone I with plus disease, stage 2 in zone II with plus disease, or stage 3 in zone II with plus disease. Type 1 ROP was defined as a treatment indication in the Early Treatment for Retinopathy of Prematurity (ETROP) study [13], requiring intervention within 24–48 h. Type 2 ROP: Stage 1 or 2 ROP in zone I, or stage 3 ROP in zone II. Type 2 ROP was classified as an observational condition in the ETROP study [13]. Vitreoretinal surgery may be

indicated for: (1) stage 4 A ROP with progressive disease demonstrating tractional retinal detachment threatening macular involvement, (2) stage 4B ROP, and (3) stage 5 ROP [14–16].

Data collection

Data collection was performed by reviewing electronic medical records of enrolled neonates during hospitalization and follow-up outpatient visits to document risk factors and monitor retinal disease progression. Based on domestic and international literature, we systematically collected information potentially associated with the development of ROP from electronic health records, including general neonatal characteristics, maternal factors, neonatal comorbidities during hospitalization, and therapeutic interventions. The specific parameters were as follows: Maternal factors (48 items): Intrauterine infection; complete placenta previa with hemorrhage; threatened preterm labor; fetal distress; intrahepatic cholestasis of pregnancy; severe eclampsia; pregnancy with mycotic vaginitis; abnormal prenatal screening; fetal distress; pregnancy with chorioamnionitis; placental abruption; premature rupture of membranes; threatened abortion; inevitable abortion; in vitro fertilization-embryo transfer; breech presentation; abnormal umbilical artery flow; fetal growth restriction; gestational diabetes mellitus; gestational hypertension; pregnancy with hypothyroidism; pregnancy with uterine fibroids; advanced maternal age; scarred uterus; cervical cerclage; twin pregnancy; cervical incompetence; mid-trimester pregnancy; heart failure; pregnancy with anemia; pregnancy with hypoproteinemia; high-risk pregnancy monitoring; adenomyosis; placenta accreta with hemorrhage; adverse obstetric history; thalassemia; hepatitis B virus carrier; ectopic pregnancy; nuchal cord; preterm premature rupture of membranes with preterm birth; hyperthyroidism; velamentous cord insertion; placental adhesion; fetal chromosomal abnormalities; severe pneumonia; electrolyte imbalance; antenatal steroid administration; blood type. Neonatal factors (55 items): Gestational age at birth; birth weight; gender; 1-minute Apgar score; 5-minute Apgar score; 10-minute Apgar score; neonatal respiratory failure; bronchopulmonary dysplasia; neonatal apnea; meconium aspiration syndrome; respiratory distress syndrome; mild neonatal asphyxia; low birth weight; neonatal pneumonia; neonatal hyperbilirubinemia; non-traumatic intraventricular hemorrhage; wet lung syndrome; patent foramen ovale; anemia of prematurity; maternal placenta previa; neonatal sepsis; test-tube baby; feeding intolerance; neonatal hyperkalemia; growth retardation; purulent meningitis; birth trauma; extremely low birth weight; transient neonatal hypothyroidism; pleural effusion; pulmonary hypertension; patent ductus arteriosus; atrial septal defect; electrolyte imbalance; acidosis; ABO hemolytic disease; neonatal hyperglycemia; thyroid dysfunction; hypoxic-ischemic myocardial injury; hypoxic-ischemic encephalopathy; hypoproteinemia; Ureaplasma urealyticum infection; iron deficiency; vitamin D deficiency; pulmonary hemorrhage; septic shock; coagulation disorders; neonatal specific infections; maternal congenital heart disease; favism; inherited metabolic disorders; necrotizing enterocolitis; congenital syphilis; multiple organ dysfunction syndrome; thalassemia. Neonatal interventions (6 items): Oxygen therapy; duration of oxygen therapy; oxygen concentration; oxygen delivery method; antibiotic administration; blood transfusion.

Application of Lasso regression in analyzing risk factors for retinopathy of prematurity

Given the high dimensionality of 109 potential risk factors observed in this study, Least Absolute Shrinkage and Selection Operator (LASSO)regression was employed to screen for significant independent predictors. In this study, the dependent variable was defined as follows: cases with immature peripheral retinal vascularization [12] were labeled as 1, while those diagnosed with ROP (regardless of type) were labeled as 0. The independent variables (risk factors) were dichotomized: presence of a risk factor was coded as 1 (confirmed diagnosis), and absence as 0 (no diagnosis). Additional coding schemes included: gender (1: male, 2: female) and delivery mode (1: cesarean section, 2: vaginal delivery).

Construction of predictive models using seven machine learning methods

The significant risk factors screened by Lasso regression were utilized as input variables. Seven predictive models were constructed using the following machine learning algorithms: RF, Extreme Gradient Boosting (XG Boost), K-Nearest Neighbors (KNN), Decision Tree (DT), Logistic Regression (LR), Light Gradient Boosting Machine (Light GBM), and Support Vector Machine (SVM). The dataset was partitioned into a training set (70% of the data) and a test set (30% of the data), with five-fold cross-validation implemented for model validation.

Observational indicators

The performance of different models was evaluated based on the following metrics: AUC, Accuracy, Precision, Sensitivity, Specificity, F1 score, Cohen's Kappa coefficient.

Statistical methods

Statistical analyses were performed using R 4.4.1 and Python 3.13.1.Continuous variables: Normally distributed data were expressed as mean ± standard deviation ($\bar{x} \pm s$), Non-normally distributed data were described using median and interquartile range (IQR), Intergroup comparisons were conducted with independent samples

t-test, Categorical variables: Expressed as absolute counts and percentages Intergroup comparisons utilized Pearson's chi-square test or Fisher's exact test. All hypothesis tests were two-sided, with $P < 0.05$ considered statistically significant.

Results

Screening results of ROP

In this study, a total of 586 neonates underwent continuous screening for 9 weeks. Among them, 63 infants developed ROP, with an overall incidence rate of 10.75% (63/586). A total of 33 infants (5.6%, 33/586) received treatment, including 33 cases treated with laser or anti-VEGF therapy, among which 2 cases underwent surgical intervention. The distribution of ROP by zones was as follows: Zone III: 33 cases (5.63%), Zone II: 19 cases (3.24%), Zone I: 10 cases (1.70%). Staging results showed: Stage 1: 33 cases (5.63%), Stage 2: 8 cases (1.37%), Stage 3: 11 cases (1.87%), Stage 4: 2 cases (0.34%). No Stage 5 ROP was observed. Additionally, plus disease (PLUS) was detected in 11 cases (1.87%), and aggressive ROP (A-ROP) occurred in 10 infants (1.71%). The detailed distribution is illustrated in Fig. 1.

Lasso regression screening of risk factors

The Lasso regression analysis identified 46 critical risk factors associated with ROP, including: Gestational age at birth, Threatened preterm labor, Fetal distress, Birth weight, Neonates born to mothers with thrombocytopenia, Neonatal hypoxic-ischemic myocardial injury, Mid-trimester pregnancy complications, Neonatal hyperkalemia, Hyperlipidemia, Mode of delivery Patent ductus arteriosus, Twin pregnancy, Neonatal hyperglycemia, Electrolyte imbalance, Neonatal hyperbilirubinemia, Septic shock, Umbilical cord entanglement, Scarred uterus, Oxygen concentration, Neonatal respiratory distress syndrome, Oxygen therapy, Neonatal necrotizing enterocolitis, Pulmonary hypertension, Premature rupture of membranes, Low birth weight, Neonatal pneumonia, Pregnancy-associated anemia, Neonatal feeding intolerance, Neonatal pulmonary hemorrhage, Antenatal corticosteroid use, Antibiotic use, In vitro fertilization and embryo transfer, Anemia in preterm infants, Pregnancy with uterine fibroids, Growth retardation, Iron deficiency, Thalassemia, Gestational diabetes mellitus, High-risk pregnancy monitoring, Pregnancy-associated mycotic vaginitis, Neonatal sepsis, Maternal electrolyte imbalance, Pregnancy-associated hypoproteinemia, Severe preeclampsia, Extremely low birth weight, The distribution and coefficients of these risk factors are illustrated in Fig. 2 – 1 and 2.

Univariate analysis of significant ROP risk factors after LASSO regression screening

Data collection focused on three categories: neonatal factors, maternal factors, and neonatal intervention measures. Following LASSO regression screening, the remaining 46 risk factors were stratified into two groups based on ROP status (ROP group vs. non-ROP group). The univariate analysis results are presented as follows: As shown in Table 1-1, analysis of continuous variables revealed statistically significant differences between groups. The ROP group demonstrated lower mean gestational age (GA) ($p < 0.001$) and lower birth weight (BW) ($p = 0.001$) compared to the non-ROP group. Notably, the ROP group required significantly higher oxygen concentrations than the non-ROP group ($p < 0.001$). These findings indicate that infants developing ROP exhibited greater developmental immaturity and required more intensive oxygen supplementation compared to their non-affected counterparts.

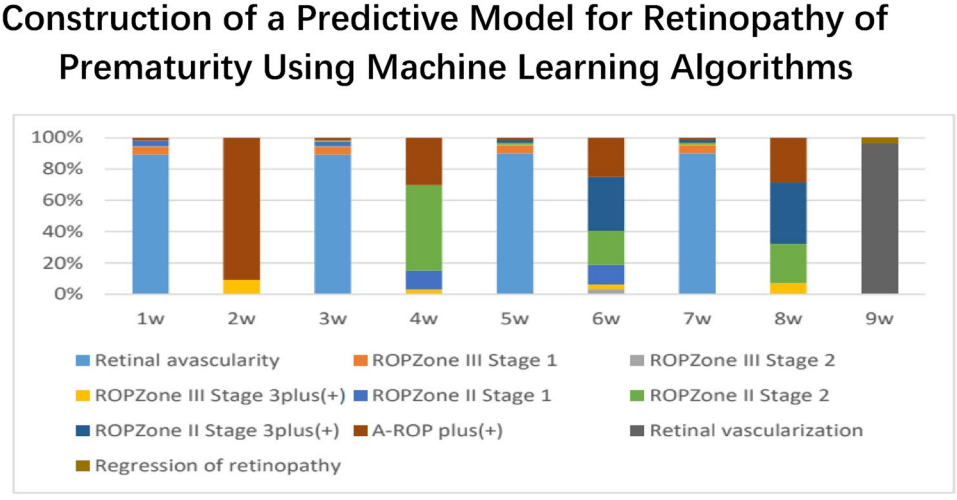


Fig. 1 Temporal progression of ROP staging and zonal distribution in 586 neonates during a 9-week ophthalmic screening protocol (%)

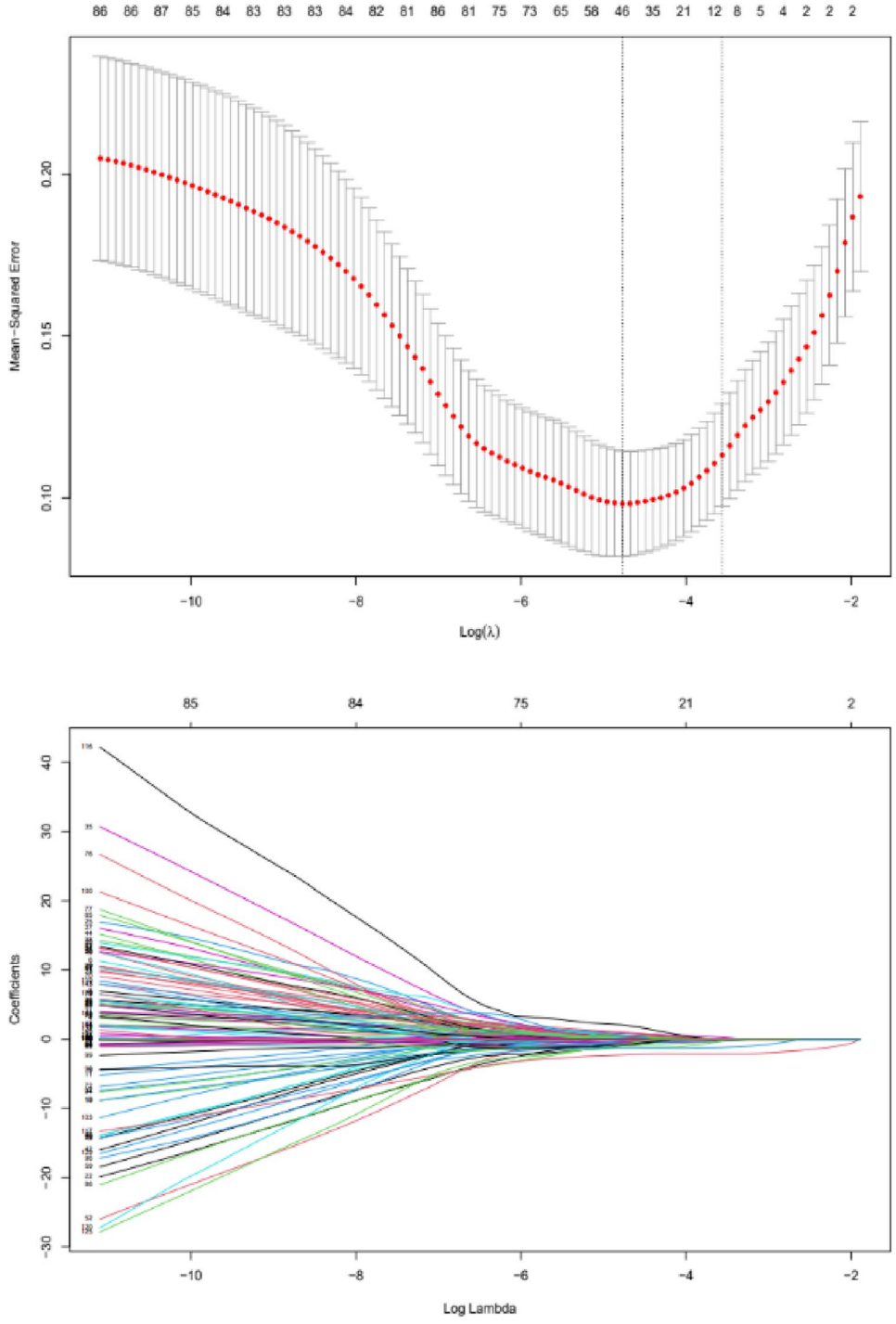


Fig. 2 1 Regularization Path of Lasso Regression Coefficients. 2 Lasso Regression Cross-Validation Curve Diagram

The results of the analysis on neonatal categorical variables are presented in Fig.3 – 1 (Refer to the following sections). Several factors showed statistically significant associations with ROP occurrence (all $p < 0.05$), including: Oxygen therapy, Neonatal respiratory distress syndrome, Blood transfusion, Low birth weight infants(LBW, BW<2500 g), Neonatal pneumonia, Umbilical cord entanglement, Neonatal hyperbilirubinemia, Premature infant anemia, Neonatal sepsis, Antibiotic usage, Placental abruption, Patent ductus arteriosus, Neonatal necrotizing enterocolitis, In vitro fertilization-embryo transfer. Conversely, the following factors demonstrated no statistically significant differences between the non-ROP and ROP groups ($p > 0.05$): Neonatal feeding intolerance,

Table 1 1. Univariate analysis of ROP risk factors (Continuous variables, Mean ± SD)

Risk Factors	non-ROP group (*n* = 523)	ROP group (*n* = 63)	*P*
GA(weeks)	31.64 ± 1.55	29.97 ± 2.44	<0.001***
BW(g)	1652.18 ± 359.02	1414.28 ± 357.76	0.001**
Oxygen Concentrations (%)	24.94 ± 4.07	34.70 ± 7.82	<0.001***

1.GA: Gestational age, BW: Birth weight 2. *$P < 0.05$, **$P < 0.01$, ***$P < 0.001$

Neonatal hyperkalemia, Growth retardation, extremely low birth weight infants(ELBW, BW<1000 g), Pulmonary hypertension, Electrolyte imbalance, Neonatal hyperglycemia, Neonatal hypoxic-ischemic myocardial injury, Neonatal pulmonary hemorrhage, Septic shock, Maternal thrombocytopenia in neonates.

The results of the categorical variable analysis for maternal factors are presented in Fig. 3 – 2 (Refer to the following sections). The following factors demonstrated statistically significant associations with ROP (all $p < 0.05$): Antenatal steroid use, Threatened preterm labor, Pregnancy with mycotic vaginitis, Gestational diabetes mellitus, Mid-trimester pregnancy, Pregnancy with anemia, Thalassemia, Pregnancy with hypoproteinemia, Maternal electrolyte imbalance. In contrast, no statistically significant differences were observed between the non-ROP and ROP groups for the following factors ($p > 0.05$): Mode of delivery, Twin pregnancy, Hyperlipidemia, Intrauterine fetal distress, Pregnancy with uterine fibroids, Scarred uterus, High-risk pregnancy supervision, Iron deficiency, Severe preeclampsia.

Machine learning algorithm for predictive model construction

Results of RF predictive model

This study employed seven metrics to evaluate model performance (Figs.4– 1 and 2, Table 2). The training set vs. test set exhibited synchronized declines in AUC (1.00 vs. 0.981), F1-score (99.7% vs. 77.4%), and Kappa coefficient (0.994 vs. 0.751), indicating slight overfitting in the test set. High-level maintenance was observed in test set accuracy (99.7% vs. 95.7%) and specificity (99.7% vs. 99.3%). Decreases in test set precision (99.7% vs. 92.3%) and sensitivity (99.7% vs. 66.7%) suggested moderate reduction in reliability of positive class predictions, potentially influenced by class imbalance. Notably, all AUC values surpassed the benchmark threshold of 0.90, confirming the model's interesting and demonstrating practical utility.

Results of XG boost predictive model

This study employed seven metrics to evaluate model performance (Figs. 5– 1 and 2, Table 3). Synchronized declines were observed between training and test sets: AUC (0.991 vs. 0.963), F1-score (99.1% vs. 81.2%), and Kappa coefficient (0.982 vs. 0.793), indicating mild overfitting in the test set. Nevertheless, the test set maintained high-level accuracy (96.3%) and specificity (99.3%). Notably, the reductions in test set precision (92.8%) and sensitivity (72.2%) suggested increased false-negative rates and diminished reliability of positive class predictions, potentially attributable to class imbalance. Crucially, all AUC values surpassed the baseline threshold of 0.9, demonstrating the model's excellent practical utility.

Results of KNN predictive model

This study employed seven metrics to evaluate model performance (Figs. 6 – 1 and 2, Table 4). The training set vs. test set exhibited marked synchronized declines in AUC (0.978 vs. 0.851), F1-score (97.9% vs. 65.1%), and Kappa coefficient (0.956 vs. 0.600), indicating significant overfitting and highlighting the need for generalization capability optimization. Despite these declines, the test set maintained relatively high accuracy (97.8% vs. 90.9%) and specificity (95.9% vs. 92.4%). Notably, pronounced reductions in test set precision (96.1% vs. 56%) and sensitivity (99.7% vs. 77.8%) suggested substantially increased false-negative rates and compromised reliability in positive class predictions, potentially attributable to class imbalance. Crucially, the AUC values remained above the baseline threshold of 0.8, demonstrating the model's retained practical utility for clinical applications.

Results of DT predictive model

This study employed seven metrics to evaluate model performance (Figs. 7 – 1 and 2, Table 5). Significant synchronized declines were observed between training and test sets across multiple metrics: AUC (0.981 vs. 0.875), F1-score (96.1% vs. 73.7%), Kappa coefficient (0.924 vs. 0.703), precision (97.0% vs. 70%), and sensitivity (95.3% vs. 77.8%), indicating test set overfitting and suggesting reduced reliability in positive class predictions. Despite these declines, the test set retained relatively high accuracy (96.2% vs. 93.9%) and specificity (97.1% vs. 95.9%). Crucially, the test set AUC remained above the baseline threshold of 0.8, demonstrating the model's robust practical utility for clinical deployment.

Results of the LR prediction model

This study employed seven metrics to evaluate model performance (Figs.8– 1 and 2, Table 6). In comparisons between the training and test sets: AUC (0.994 vs. 0.926), F1-score (97.6% vs. 71.1%), Kappa coefficient (0.950 vs. 0.672), precision (96% vs. 75%), and sensitivity (99.1% vs. 66.7%) all exhibited concurrent significant declines, indicating an increased risk of overfitting and a marked reduction in the reliability of positive class predictions.

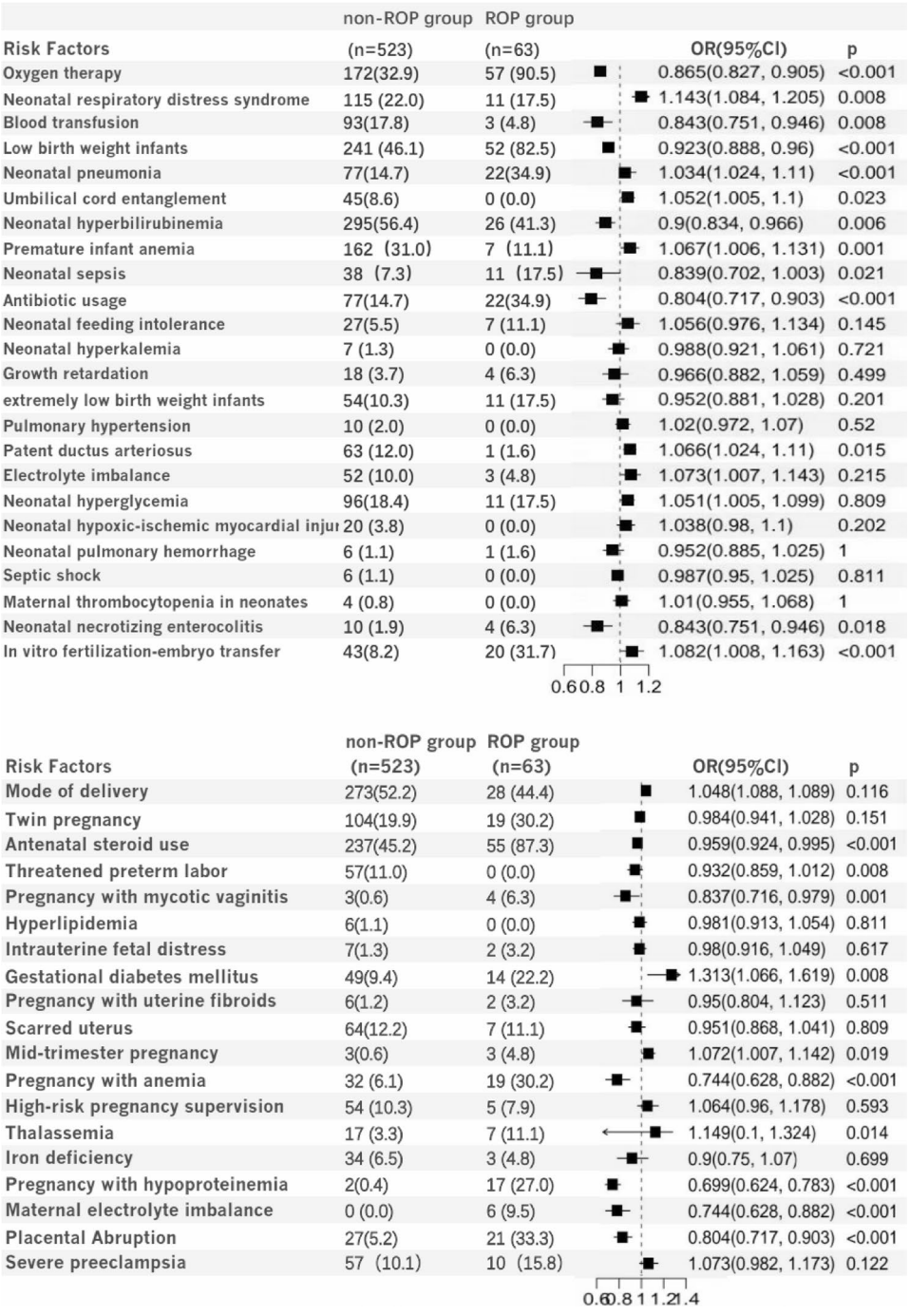


Fig. 3 1 Univariate Analysis of ROP Risk Factors in Neonates (Categorical Variables, n (%)). 2 Univariate Analysis Results of Maternal Risk Factors (Categorical Variables) for ROP (n (%))

Nevertheless, accuracy (97.5% vs. 93.9%) and specificity (95.9% vs. 91.2%) remained at high levels. Furthermore, the AUC exceeded the benchmark value of 0.9, demonstrating that the model possesses excellent practical utility.

Results of the light GBM prediction model

This study employed seven metrics to evaluate model performance (Figs. 9–1 and 2, Table 7). Comparisons between the training and test sets revealed significant declines in AUC (0.995 vs. 0.929), F1-score (96.5% vs. 68.3%), Kappa coefficient (0.930 vs. 0.638), precision (95.4% vs. 60.9%), and sensitivity (97.7% vs. 77.8%), indicating an elevated risk of overfitting and a further

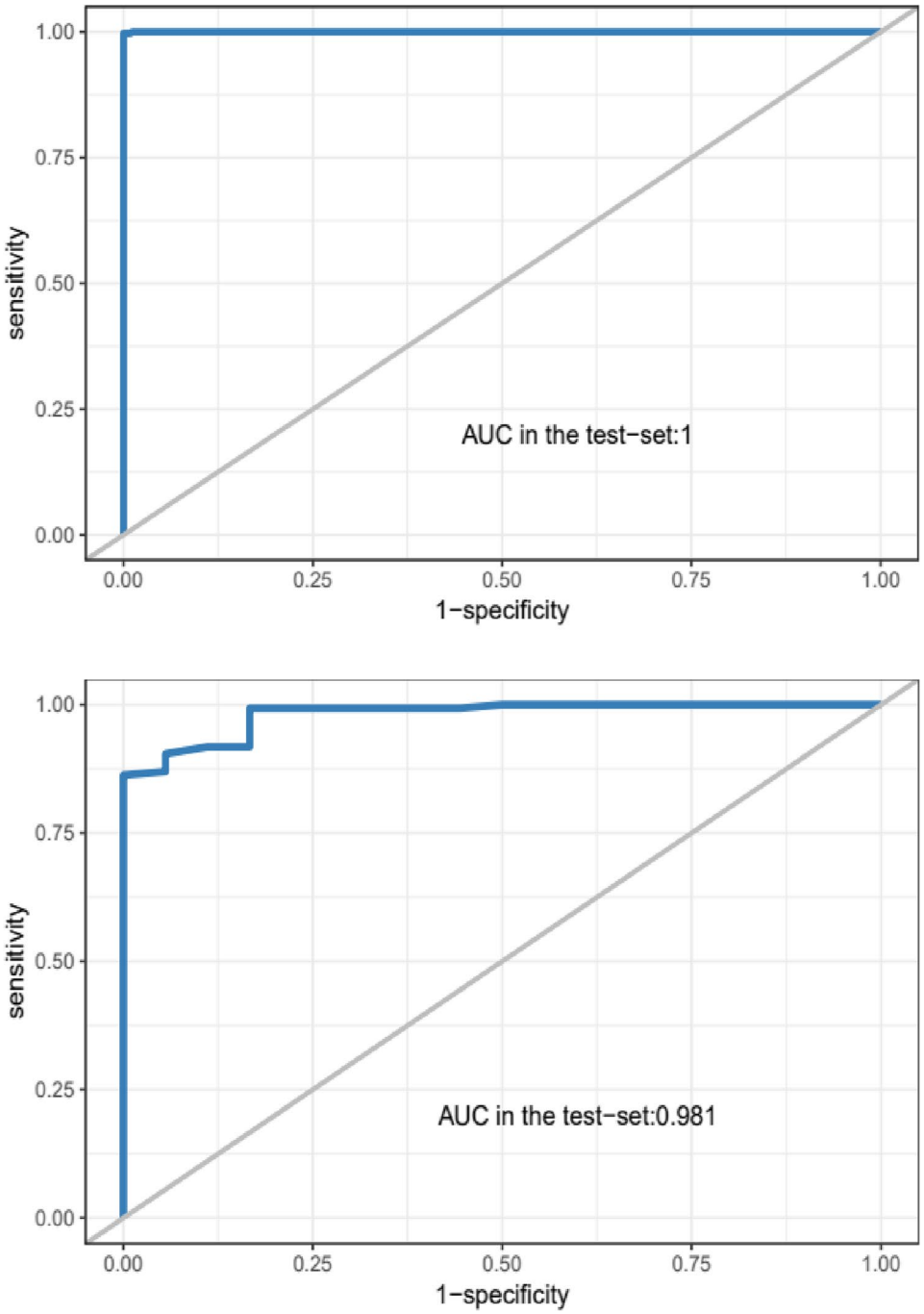


Fig. 4 1 AUC for RF Training Set. 2 AUC for RF Test Set

Table 2 Results of RF predictive model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
RF- The training set	1.000	0.997	0.997	0.997	0.997	0.997	0.994
RF-test set	0.981	0.957	0.923	0.993	0.667	0.774	0.751

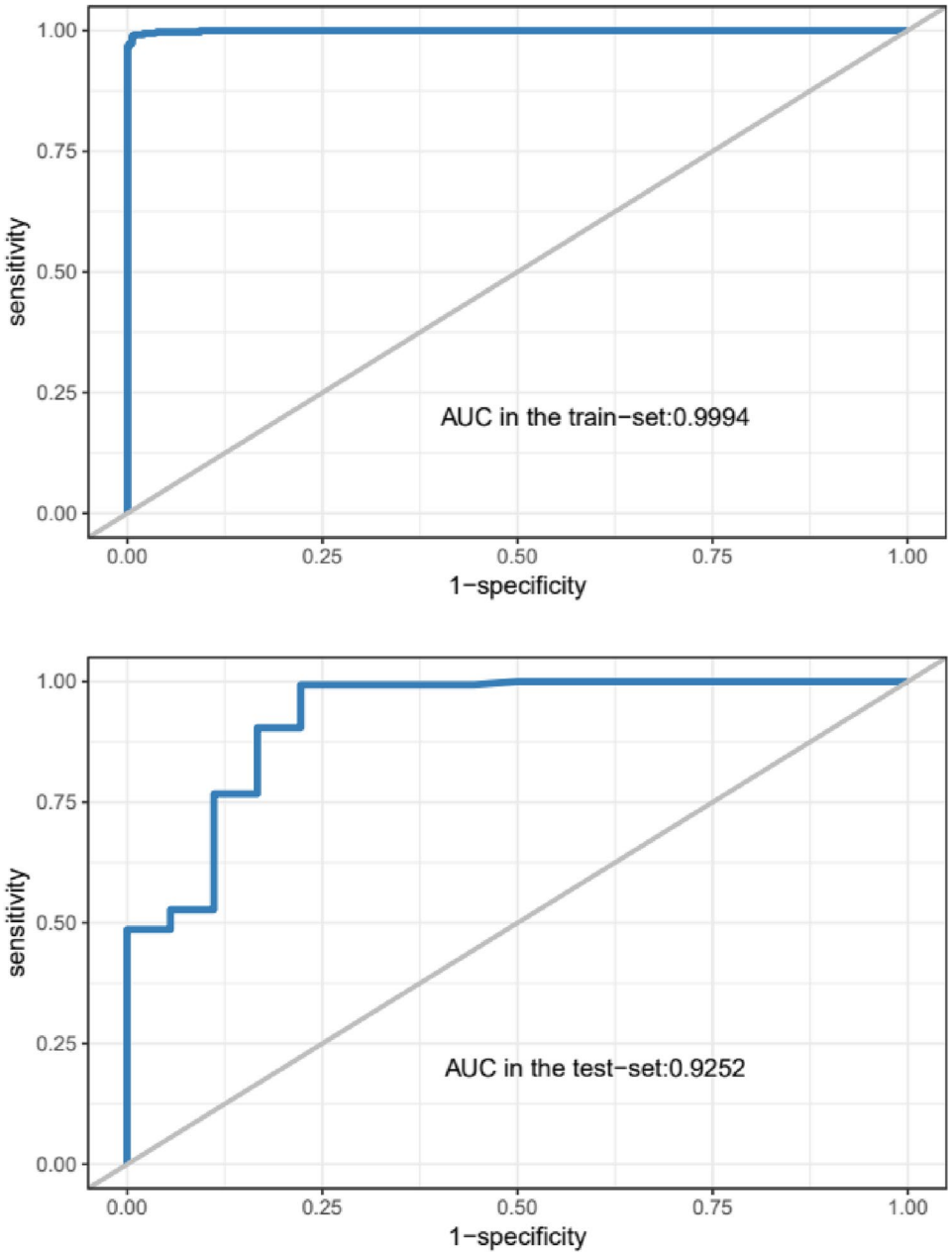


Fig. 5 1 AUC for XG Boost Training Set. 2 AUC for XG Boost Test Set

Table 3 Results of XG boost predictive model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
XG Boost-The training set	0.999	0.991	0.988	0.988	0.994	0.991	0.982
XG Boost - test set	0.925	0.963	0.928	0.993	0.722	0.812	0.793

increase in the missed detection rate for positive class samples. Nevertheless, the test set still maintained high accuracy (92.1%) and specificity (93.8%). Moreover, the AUC exceeded the benchmark value of 0.90, demonstrating the model's strong practical utility.

Results of the SVM prediction model

This study employed seven metrics to evaluate model performance (Figs. 10–1and 2, Table 8). Significant concurrent declines were observed in AUC (0.987 vs. 0.882), F1-score (98.7% vs. 82.4%), and Kappa coefficient (0.974 vs. 0.803) between the training and test sets, indicating an increased risk of overfitting. Notably, precision (98.8%

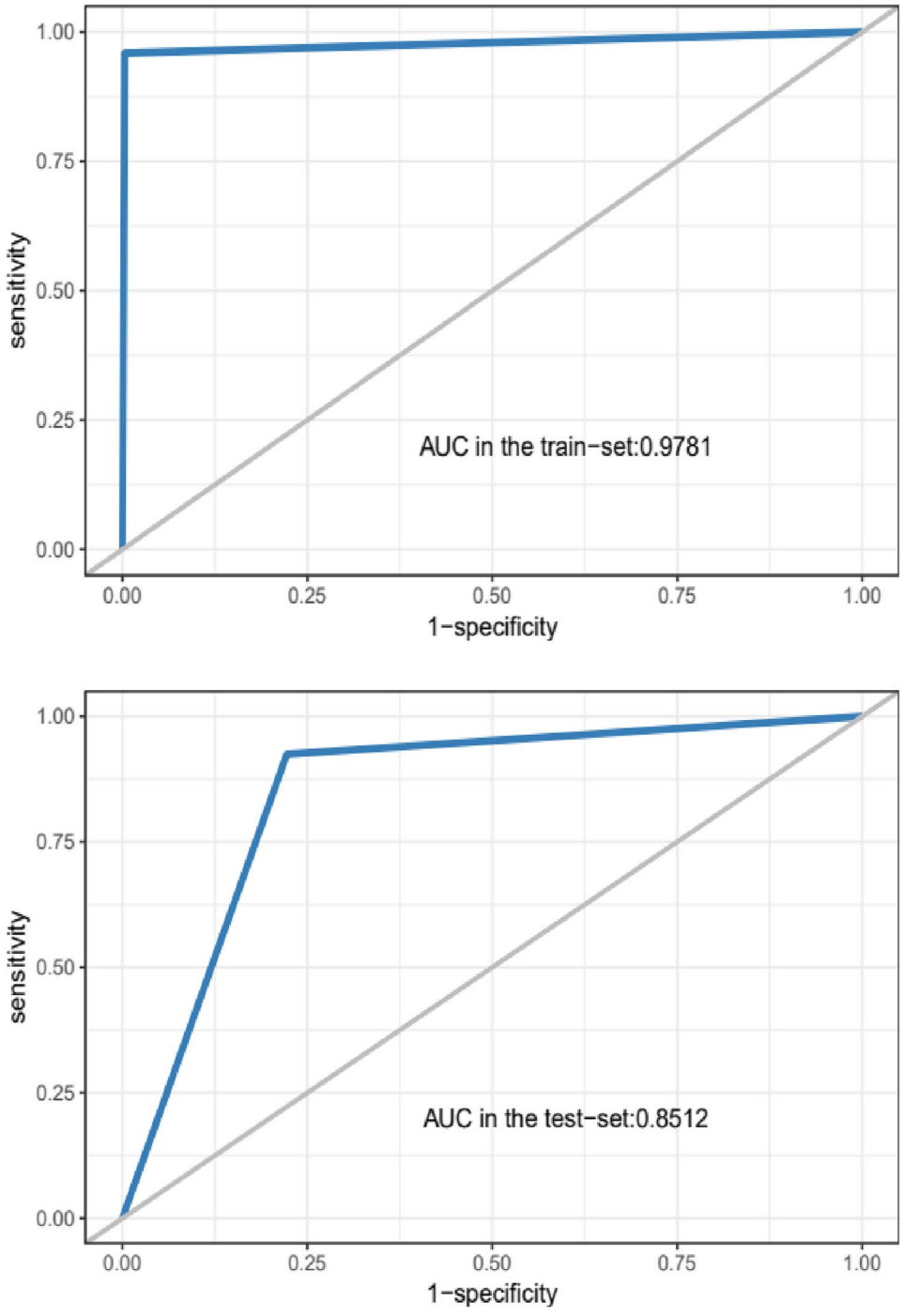


Fig. 6 1 AUC for KNN Training Set. 2 AUC for KNN Test Set

Table 4 Results of KNN predictive model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
KNN-The training set	0.978	0.978	0.961	0.959	0.997	0.979	0.956
KNN- test set	0.851	0.909	0.56	0.924	0.778	0.651	0.6

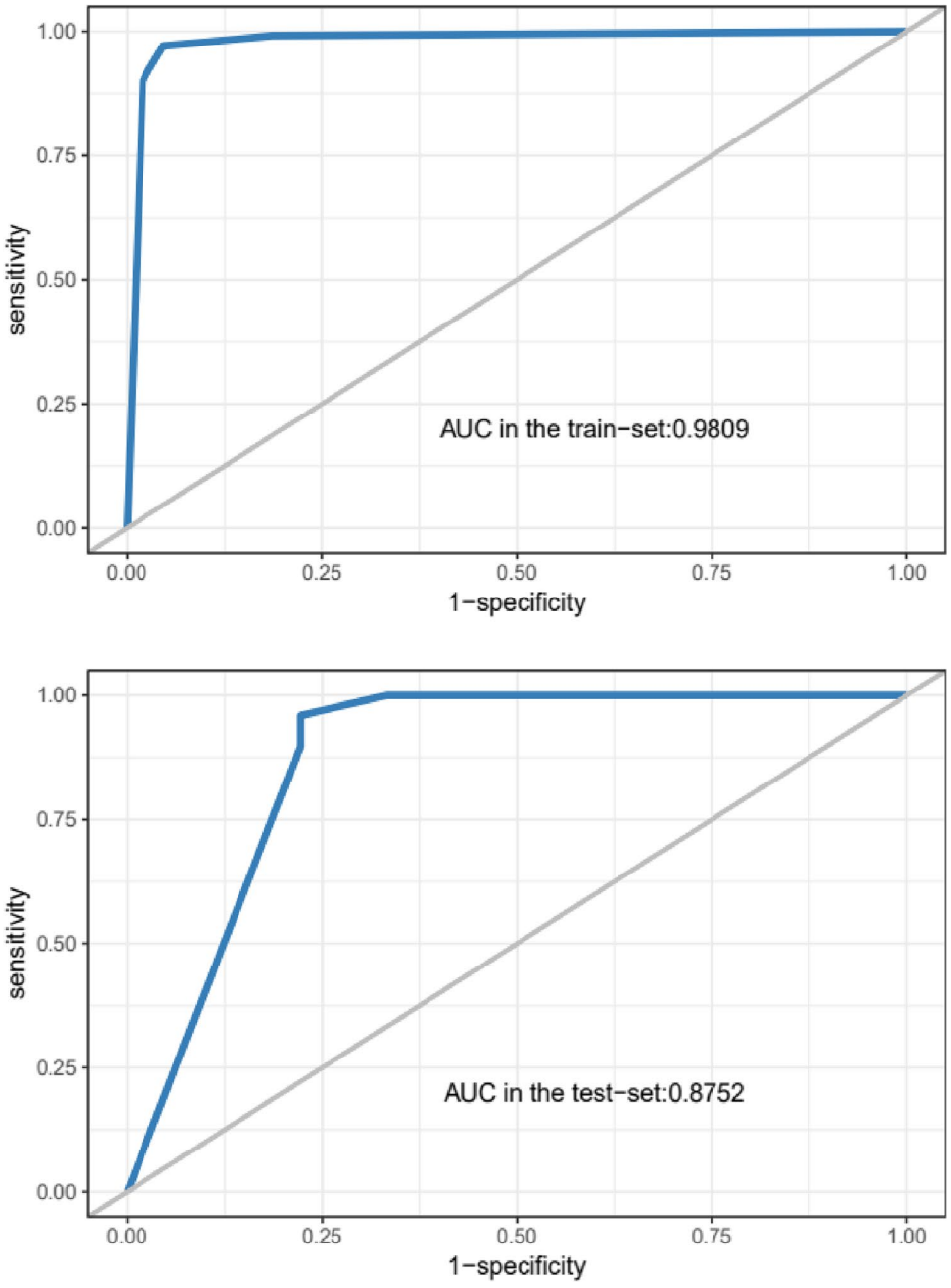


Fig. 7 1 AUC for DT Training Set. 2. AUC for DT Test Set

Table 5 Results of DT model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
DT-The training set	0.981	0.962	0.970	0.971	0.953	0.961	0.924
DT-test set	0.875	0.939	0.7	0.959	0.778	0.737	0.703

vs. 87.5%) and sensitivity (98.5% vs. 77.8%) showed substantial reductions in the test set, suggesting a marked decline in the reliability of positive class predictions. Nevertheless, the test set retained high accuracy (96.3%) and specificity (98.6%). Furthermore, the AUC surpassed the benchmark value of 0.8, demonstrating that the model retains satisfactory practical utility.

Comparison of seven machine learning models

This study evaluated the performance of seven models using seven metrics (Table 9, Table 10). Overall, the

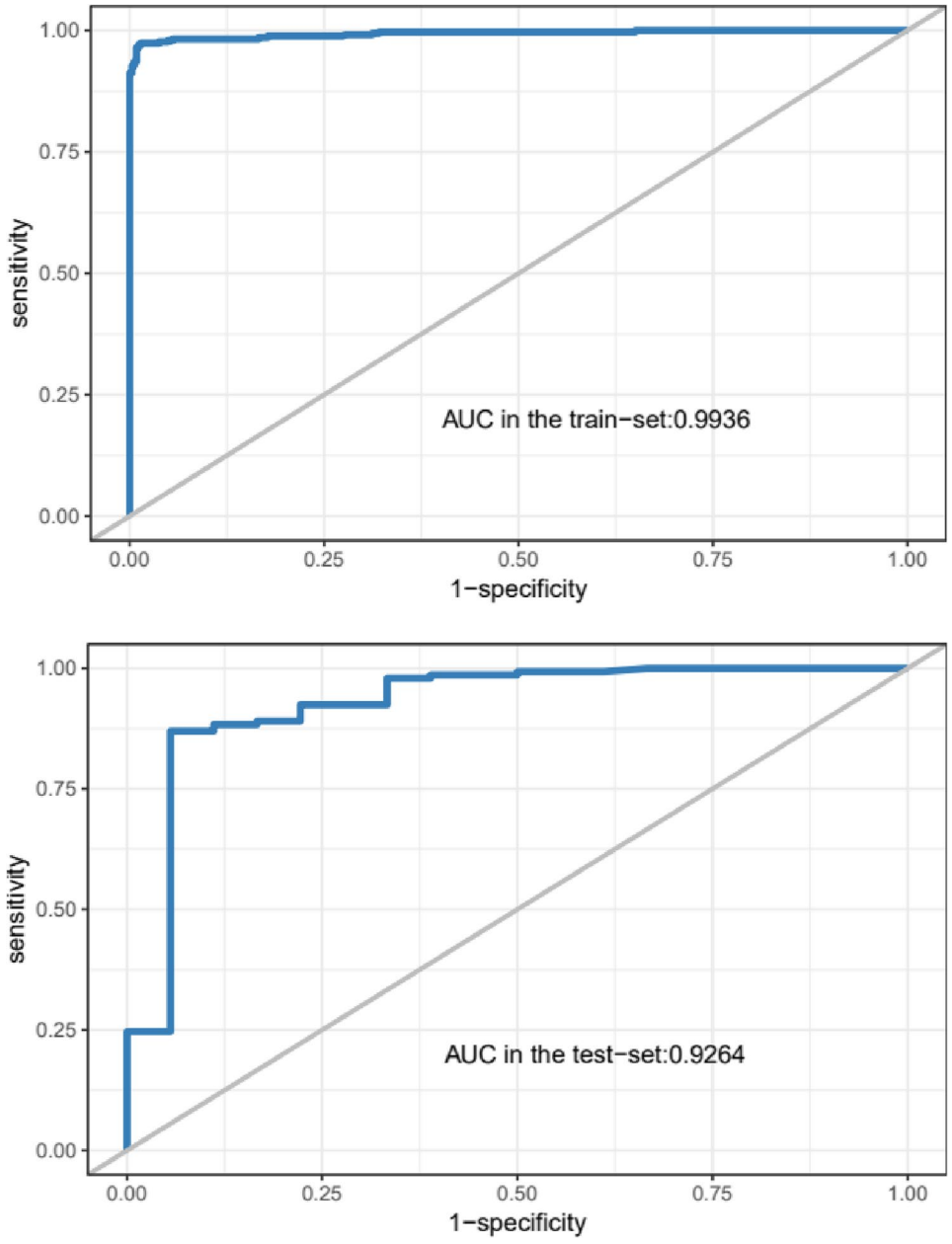


Fig. 8 1 AUC for LR Training Set. 2 AUC for LR Test Set

Table 6 Results of LR model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
LR- The training set	0.994	0.975	0.960	0.959	0.991	0.976	0.950
LR- test set	0.926	0.939	0.75	0.912	0.667	0.771	0.672

models demonstrated good consistency in performance between the training and test sets, indicating strong generalization capability. Although the test set metrics were slightly lower than those of the training set, the overall performance remained high, suggesting robust predictive ability on unseen data and satisfactory practical utility. Among the seven models, the RF and XG Boost models achieved the largest AUC values in both training and test sets, outperforming the other five machine learning models. To identify the very good model, we compared RF and XG Boost using Decision Curve Analysis (DCA) (Fig. 11). The results revealed that the RF model surpassed XG Boost, demonstrating the best overall performance among all models.

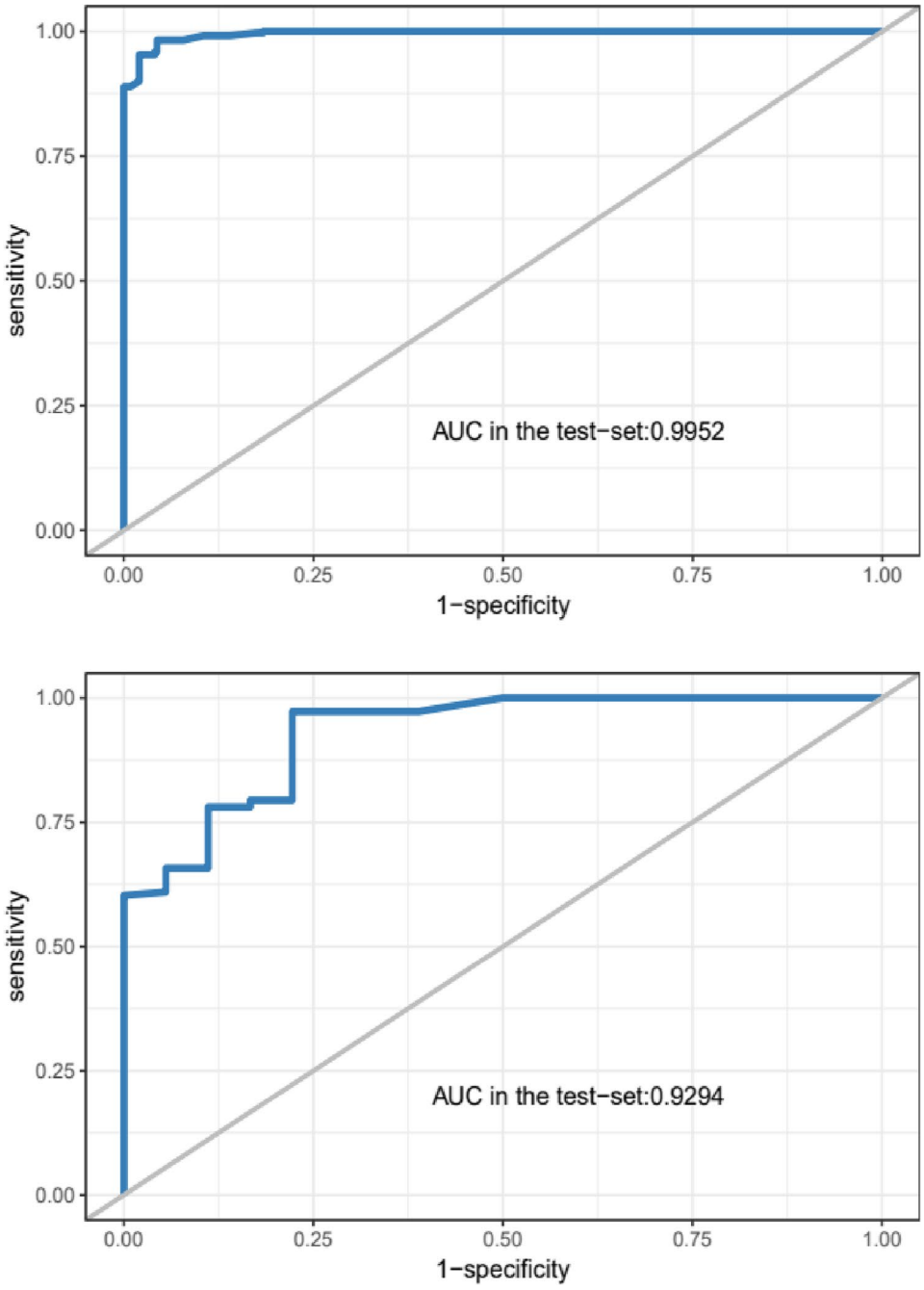


Fig. 9 1 AUC for Light GBM Training Set. 2 AUC Light GBM Test Set

Table 7 Results of light GBM model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
Light GBM-The training set	0.995	0.965	0.954	0.953	0.977	0.965	0.930
Light GBM - test set	0.929	0.921	0.609	0.938	0.778	0.683	0.638

Discussion

Clinical characteristics and ROP

The incidence of ROP in this study (10.75%) was lower than the global average in preterm infants (15–20%) [2, 3, 5], which may be attributed to the relatively lower proportion of extremely preterm infants in our cohort (mean gestational age: 31.51 weeks). Notably, three ROP cases with birth weights > 2.5 kg were identified, including one infant weighing 3.30 kg (gestational age: 31 weeks + 5 days), diverging from traditional risk factor paradigms.

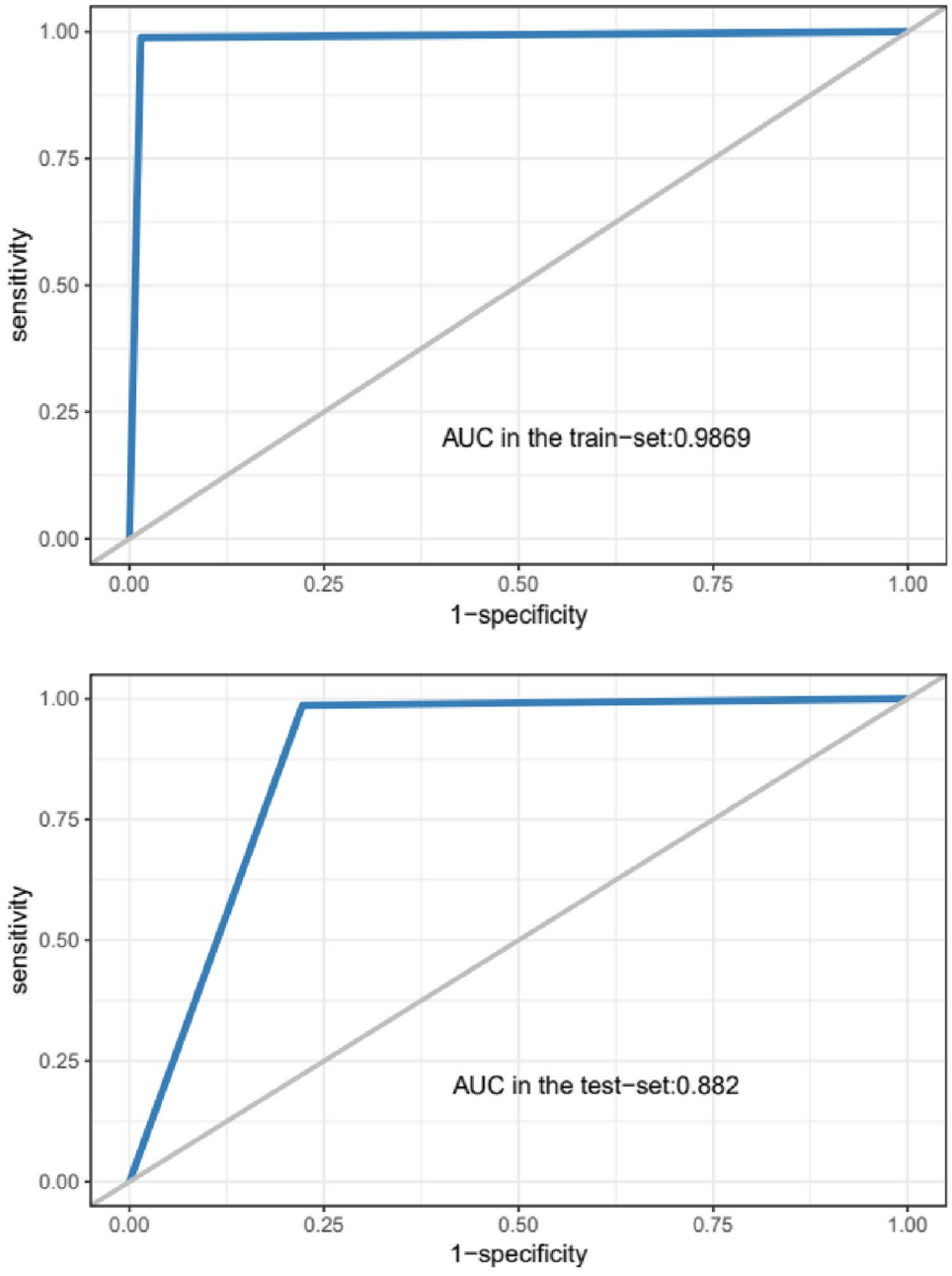


Fig. 10 1 AUC for SVM Training Set. 2 AUC for SVM Test Set

Table 8 Results of SVM model

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
SVM-The training set	0.987	0.987	0.988	0.988	0.985	0.987	0.974
SVM - test set	0.882	0.963	0.875	0.986	0.778	0.824	0.803

In developed countries, advanced neonatal intensive care has improved survival rates among very low birth weight (VLBW, BW < 1 500 g) infants but concurrently increased ROP incidence [17]. However, standardized screening protocols in developed nations have significantly reduced blindness rates through early interventions. Rekha et al. reported [18] in a prospective study of 100 premature infants with ROP that the incidence of ROP was 47.3% in preterm infants with birth weight < 1500 g, while it could be as high as 73.3% in those < 1000 g. The incidence of ROP was 83%, 60%, and 50% in preterm infants with

Table 9 Statistical results of seven machine learning algorithms on the training set

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
RF	1.000	0.997	0.997	0.997	0.997	0.997	0.994
XG Boost	0.999	0.991	0.988	0.988	0.994	0.991	0.982
KNN	0.978	0.978	0.961	0.959	0.997	0.979	0.956
DT	0.981	0.962	0.970	0.971	0.953	0.961	0.924
LR	0.994	0.975	0.960	0.959	0.991	0.976	0.950
Light GBM	0.995	0.965	0.954	0.953	0.977	0.965	0.930
SVM	0.987	0.987	0.988	0.988	0.985	0.987	0.974

Table 10 Statistical results of seven machine learning algorithms on the test set

Predictive Model	AUC	Accuracy	Precision	specificity	sensitivity	F1-score	Kappa coefficient
RF	0.981	0.957	0.923	0.993	0.667	0.774	0.751
XG Boost	0.925	0.963	0.928	0.993	0.722	0.812	0.793
KNN	0.851	0.909	0.56	0.924	0.778	0.651	0.6
DT	0.875	0.939	0.7	0.959	0.778	0.737	0.703
LR	0.926	0.939	0.75	0.912	0.667	0.771	0.672
Light GBM	0.929	0.921	0.609	0.938	0.778	0.683	0.638
SVM	0.882	0.963	0.875	0.986	0.778	0.824	0.803

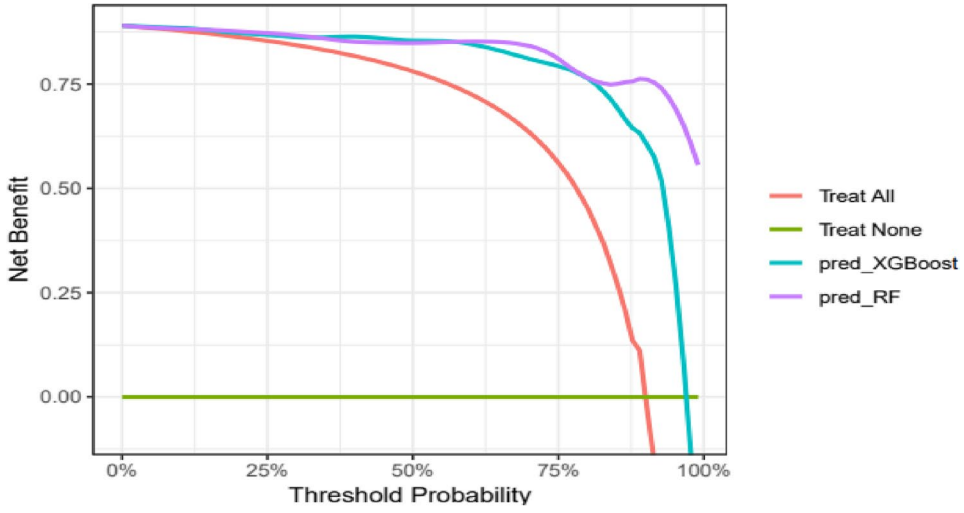


Fig. 11 DCA Comparison Between RF and XG Boost Models

gestational ages of 28–29 weeks, ~31 weeks, and ~33 weeks, respectively.

In contrast, our study highlights a distinct ROP profile in China, characterized by larger birth weights and older gestational ages compared to other countries. The maximum gestational age in our cohort was 35 weeks + 1 day, aligning with reports from other developing countries (e.g., Vietnam) [19, 20]. This phenomenon suggests that risk factors beyond gestational age and birth weight—such as oxygen therapy duration, antenatal steroid use, or postnatal care practices—may require comprehensive evaluation to refine ROP screening and prevention strategies.

The lesion distribution in this study was predominantly located in Zone III (52.38%), and the majority of cases were classified as Stage 1 (52.38%), which aligns with the retinal vascularization pattern in preterm infants. However, the identification of 10 cases of AP-ROP (1.71%) highlights the clinical necessity for establishing rapid response mechanisms. Current screening criteria vary globally: American guidelines emphasize GA ≤ 30 weeks and BW ≤ 1500 g as high-risk factors for ROP [21], while China follows the *Chinese Guidelines for Retinopathy of Prematurity Screening (2014)* [11], recommending screening for infants with BW < 2000 g or GA < 32 weeks. Although these guidelines demonstrate high Sensitivity for detecting ROP requiring treatment, they may still overlook certain high-risk cases, such as the 35 weeks + 1 day GA infant identified in this study. To address this limitation, we propose the adoption of dynamic screening criteria that integrate individualized risk factors, including sepsis, oxygen

supplementation duration, and comorbidities, for comprehensive evaluation.

Analysis of ROP-Related risk factors

In this study, Lasso regression analysis identified 46 independent risk factors for ROP. Among these, higher GA and greater BW exhibited strong protective effects against ROP development, consistent with previous studies [22]. Specifically, each 1-week increase in GA or 1-unit increase in BW was associated with a significant reduction in ROP risk. This may be attributed to the immature retinal vascularization in preterm infants with lower GA/BW, rendering them more susceptible to abnormal neovascularization under external stressors. Notably, threatened preterm birth and fetal distress were inversely correlated with ROP risk. This observation might reflect enhanced clinical surveillance in high-risk pregnancies, including earlier initiation of antenatal corticosteroid therapy and neonatal intensive care, which could indirectly mitigate oxidative stress damage. However, limited literature supports this hypothesis, necessitating validation through prospective cohort studies.

The strong positive correlation between oxygen therapy-related indices (oxygen administration and concentration) and ROP highlights the double-edged sword effect of oxygen therapy. Postnatal administration of high-concentration oxygen therapy in preterm infants serves as a critical trigger for ROP pathogenesis. Oxygen exposure suppresses normal vascular development in immature retinas, leading to vascular occlusion and subsequent abnormal neovascularization. Notably, prolonged duration and elevated concentration of oxygen therapy are dose-dependently associated with increased ROP risk [23].

This study quantified the independent effects of oxygen therapy parameters using Lasso regression analysis and proposes the clinical implementation of dynamic SpO_2 monitoring protocols (targeting 90–95% SpO_2) to avoid hyperoxia. Neonatal respiratory distress syndrome (NRDS) [24] and necrotizing enterocolitis (NEC) [23] exhibited positive correlations with ROP incidence, suggesting that systemic inflammatory responses may exacerbate retinal pathology via the IL-6/TNF-α pathway. These findings provide a theoretical foundation for exploring anti-inflammatory therapies in ROP prevention. Pulmonary hypertension and neonatal anemia aggravated retinal hypoxia by impairing oxygen delivery, thereby activating hypoxia-inducible factor (HIF) and VEGF signaling pathways [25, 26]. HIF, a key transcriptional regulator of hypoxia response comprising HIF-1α and HIF-2α subunits, is highly expressed in avascular retinal zones during early hypoxia, promoting VEGF release and pathologic angiogenesis. Subsequent hyperoxia-induced HIF degradation transiently reduces VEGF levels, but recurrent hypoxia triggers VEGF rebound, driving aberrant vascular proliferation and ROP progression. Notably, septic shock and neonatal sepsis directly stimulate abnormal angiogenesis through inflammatory cytokines (e.g., VEGF) [27]. Transfusion-associated risks may arise from HBF-to-HBA substitution, hemodynamic fluctuations, and iron overload-induced oxidative damage, necessitating vigilant fundus monitoring in infants receiving multiple transfusions. A study demonstrated [28]: a 2.77-fold increased ROP risk after ≥ 3 transfusions.

Anemia during pregnancy [29] and hypoproteinemia were positively correlated with ROP, both of which may reflect maternal malnutrition or systemic inflammatory states. These conditions severely compromise placental function, leading to fetal developmental abnormalities or preterm birth through insufficient nutrient supply. Additionally, maternal systemic inflammation and oxidative stress in severe preeclampsia [30] may transmit to the fetus via the placenta, exacerbating retinal vascular injury and pathological neovascularization. Elevated levels of soluble vascular endothelial growth factor receptor-1 (sVEGFR-1) released by preeclamptic placentas could inhibit fetal retinal VEGF signaling pathways, disrupting normal vascular development and collectively contributing to ROP pathogenesis. Notably, antenatal corticosteroid use [31] and antibiotic administration [32] exhibited positive associations with ROP in our model. This correlation may reflect the inherently higher disease severity in neonates requiring these interventions, underscoring the need for clinicians to balance therapeutic benefits against potential ROP risks. Furthermore, the significant association between in vitro fertilization-embryo transfer (IVF-ET) [33] and the development of ROP suggests that assisted reproductive technology (ART) may influence retinal development through epigenetic modifications such as DNA methylation and histone modification. However, no significant correlation was observed between twin pregnancy and ROP incidence, potentially due to limited sample sizes in current studies; thus, further investigations with expanded cohorts are warranted.

Machine Learning-Based predictive models

Our analysis revealed distinct applicability scenarios for different machine learning algorithms:The RF demonstrated superior performance in high-dimensional feature spaces with moderate sample sizes, particularly when feature importance analysis was required. XG Boost and Light GBM exhibited optimal accuracy in large-scale structured datasets, aligning with their established efficacy in data competition scenarios demanding high-precision predictions. The DT proved most effective for simple classification tasks requiring strong interpretability, though limited in handling complex non-linear relationships. The KNN and SVM were preferentially

applicable to small-sample datasets, consistent with their sensitivity to data scale. The LR remained the baseline choice for linearly separable problems due to its computational efficiency and parametric transparency.

The RF algorithm has emerged as a cornerstone in machine learning due to its efficiency, flexibility, and robustness. Its core advantage lies in reducing model variance through randomized strategies, making it particularly suitable for complex data scenarios involving large datasets and high-dimensional problems. The RF excels in providing critical outputs such as variable importance rankings and similarity matrices, while demonstrating resistance to overfitting, rapid training speeds, and superior classification performance. These attributes have cemented its widespread application in classification, regression, and feature selection tasks.

However, the RF model exhibits limitations, including elevated computational costs and reduced interpretability due to its inherent "black-box" nature [34]. In this study, the RF model architecture outperformed six other machine learning models in structural optimization. Nevertheless, the observed decline in test-set sensitivity (from 92.4 to 85.1%) and F1-score (from 0.91 to 0.83) compared to training metrics may stem from insufficient sample size ($n = 680$) and excessive inclusion of risk factors (46 variables). To enhance model generalizability, we propose expanding the cohort size in subsequent phases and validating findings through external clinical datasets.

Conclusion

Through the comparison of seven machine learning models constructed based on 46 critical risk factors associated with ROP to predict disease occurrence, the RF model demonstrated superior performance over the other six models across multiple metrics in both training and testing sets. Specifically, the RF model achieved higher values in AUC, accuracy, precision, specificity, F1-score, and Kappa coefficient. These results indicate that the RF model exhibits strong predictive value for ROP, providing a clinically useful approach to early identification of high-risk populations for ROP.

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Authors' contributions
YY collected, analyzed and interpreted the patient data, YY was a major contributor in writing the manuscript. LL, YL, LS, HZ, RS and PF performed the ocular examination. YY performed data processing and analysis. MH conceived the study and supervised the research. All authors read and approved the final manuscript.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study adhered to the principles of the Declaration of Helsinki and was approved by the Medical Ethics Committee of the First Affiliated Hospital of Guangxi Medical University (Ethics Approval No.: 2024-K047-01), which waived the requirement for written informed consent.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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